

Keshav Rajput

New York, NY | keshav.rajput4@gmail.com | +1 646 299 9431
github.com/happycoder | linkedin.com/in/keshav-rajput-862267259 | happycoder.github.io

EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Design and Analysis of Algorithms II, Principles of Database Systems, Open Source and Professional Software Development, Distributed Systems, Operating Systems, Computer Networks

PROJECTS

Atlas | C++23, SDL_GPU, HLSL, WebAssembly, CMake, vcpkg github.com/happycoder/atlas-engine, 2026

- C++23 engine for map-based strategy games: SDL_GPU renderer, asset pipeline with hot reload, versioned scene serialization.
- Deterministic fixed-tick simulation with state hashing, replay and lockstep; mods run sandboxed in WebAssembly.
- CI with sanitizers, lint and GPU smoke tests; benchmarks gated at 1.25x of a recorded baseline; 16 ADRs.

mlds-auth | C11, liboqs, libsodium, SQLite, CMake, libFuzzer github.com/happycoder/mlds-auth, 2026

- Post-quantum mutual authentication and session protocol in C11: ML-DSA-65 signatures over hybrid X25519 + ML-KEM-768.
- 0.45 ms median handshake; ChaCha20-Poly1305 record layer at 736 MiB/s.
- 11 libFuzzer targets, 169 mutation tests, ASan/UBSan CI on Linux and macOS; login daemon with device enrolment and revocation.

ForgeLog | Kotlin, Jetpack Compose, Room, Hilt, Robolectric github.com/happycoder/ForgeLog, 2026

- Offline Android gym tracker in Kotlin: Jetpack Compose, Room, Hilt, MVVM. Declares no network permission.
- Rest timer survives process death; analytics for volume, e1RM and personal bests; JSON backup and CSV export.
- 830 tests that run without a device; signed 2.3 MB release APK.

Custodian | TypeScript, MCP, OAuth 2.1, Preact, AWS CDK, DynamoDB github.com/happycoder/custodian, 2026

- MCP server and Alexa+ add-on that checks a household inventory against CPSC, NHTSA and FDA recall feeds. Built for Amazon Build, Ship, Shape 2026.
- Wrote the OAuth 2.1 authorization server: PKCE S256, rotating refresh tokens.
- TypeScript monorepo on AWS CDK (App Runner, DynamoDB); 10 tools, 78 tests, tool calls inside a 500 ms budget.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952, about top 1.2% of 3,594 teams.
- **Amazon Build, Ship, Shape 2026 (Alexa+ track)**, 2026. Shipped Custodian, an MCP recall-monitoring add-on.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash

Systems: CMake, gRPC, SQLite, WebAssembly, Android (Jetpack Compose, Room), Node.js, REST APIs, MCP

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Linux, LaTeX

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

ACTIVITIES

- Treasurer, OWASP Student Chapter, IIIT Delhi